Code quality

Introduction

What

> Always code as if the person who ends up maintaining your code will be a violent psychopath who knows where you live. -- Martin Golding

Production code needs to be of high quality. Given how the world is becoming increasingly dependent on software, poor quality code is something no one can afford to tolerate.

Guideline: Maximize readability

Introduction

> Programs should be written and polished until they acquire publication quality. --Niklaus Wirth

Among various dimensions of code quality, such as run-time efficiency, security, and robustness, one of the most important is readability (aka understandability). This is because in any non-trivial software project, code needs to be read, understood, and modified by other developers later on. Even if you do not intend to pass the code to someone else, code quality is still important because you will become a 'stranger' to your own code someday.

Basic

Avoid long methods

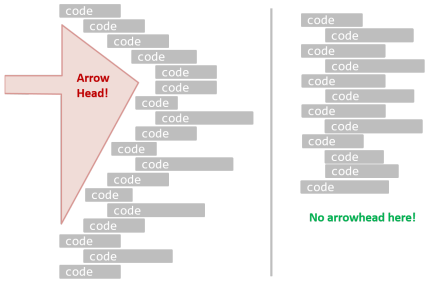
Avoid long methods as they often contain more information than what the reader can process at a time. Consider if shortening is possible when a method goes beyond 30 LoC. The bigger the haystack, the harder it is to find a needle.

Avoid deep nesting

> If you need more than 3 levels of indentation, you're screwed anyway, and should fix your program. --Linux 1.3.53 Coding Style

Avoid deep nesting -- the deeper the nesting, the harder it is for the reader to keep track of the logic.

In particular, avoid *arrowhead* style code.



A real code example:

Bad

```
int subsidy() {
    int subsidy;
    if (!age) {
        if (!sub) {
            if (!notFullTime) {
                subsidy = 500;
            } else {
                subsidy = 250;
            }
        } else {
            subsidy = 250;
        }
    } else {
        subsidy = -1;
    }
    return subsidy;
}
```

Good

```
int calculateSubsidy() {
    int subsidy;
    if (isSenior) {
        subsidy = REJECT_SENIOR;
    } else if (isAlreadySubsidized) {
        subsidy = SUBSIDIZED_SUBSIDY;
    } else if (isPartTime) {
        subsidy = FULLTIME_SUBSIDY * RATIO;
    } else {
        subsidy = FULLTIME_SUBSIDY;
    }
    return subsidy;
}
```

Avoid complicated expressions

Avoid complicated expressions, especially those having many negations and nested parentheses. If you must evaluate complicated expressions, have it done in steps (i.e., calculate some intermediate values first and use them to calculate the final value).

Bad

```
return ((length < MAX_LENGTH) || (previousSize != length))
        && (typeCode == URGENT);
```

Good

```
boolean isWithinSizeLimit = length < MAX_LENGTH;
boolean isSameSize = previousSize != length;
boolean isValidCode = isWithinSizeLimit || isSameSize;

boolean isUrgent = typeCode == URGENT;

return isValidCode && isUrgent;
```

> ❝ The competent programmer is fully aware of the strictly limited size of his own skull; therefore he approaches the programming task in full humility, and among other things he avoids clever tricks like the plague. ❞ -- Edsger Dijkstra

Avoid magic numbers

Avoid *magic numbers* in your code. When the code has a number that does not explain the meaning of the number, it is called a "magic number" (as in "the number appears as if by magic"). Using a *named constant* makes the code easier to understand because the name tells us more about the meaning of the number.

Bad

```
return 3.14236;
...
return 9;
```

Good

```
static final double PI = 3.14236;
static final int MAX_SIZE = 10;
...
return PI;
...
return MAX_SIZE - 1;
```

Similarly, you can have 'magic' values of other data types.

Bad

```
return "Error 1432"; // A magic string!
```

Avoid any *magic literals* in general, not just magic numbers.

Make the code obvious

Make the code as explicit as possible, even if the language syntax allows them to be implicit. Here are some examples:

- [`Java`] Use explicit type conversion instead of implicit type conversion.
- [`Java` , `Python`] Use parentheses/braces to show groupings even when they can be skipped.
- [`Java` , `Python`] Use enumerations when a certain variable can take only a small number of finite values. For example, instead of declaring the variable 'state' as an integer and using values 0, 1, 2 to denote the states 'starting', 'enabled', and 'disabled' respectively, declare 'state' as type `SystemState` and define an enumeration `SystemState` that has values `'STARTING'` , `'ENABLED'` , and `'DISABLED'` .

Intermediate

Structure code logically

Lay out the code so that it adheres to the logical structure. The code should read like a story. Just like how you use section breaks, chapters and paragraphs to organize a story, use classes, methods, indentation and line spacing in your code to group related segments of the code. For example, you can use blank lines to separate groups of related statements.

Sometimes, the correctness of your code does not depend on the order in which you perform certain intermediary steps. Nevertheless, this order may affect the clarity of the story you are trying to tell. Choose the order that makes the story most readable.

Bad

```
statement A1
statement A2
statement A3
statement B1
statement C1
statement B2
statement C2
```

Good

```
statement A1
statement A2
statement A3

statement B1
statement B2

statement C1
statement C2
```

Do not 'Trip Up' reader

Avoid things that would make the reader go 'huh?', such as,

- unused parameters in the method signature
- similar things that look different
- different things that look similar
- multiple statements in the same line
- data flow anomalies such as, pre-assigning values to variables and modifying it without any use of the pre-assigned value

Practice KISSing

Do not try to write 'clever' code. "Keep it simple, stupid" (KISS), as the old adage goes. For example, do not dismiss the brute-force yet simple solution in favor of a complicated one because of some 'supposed benefits' such as 'better reusability' unless you have a strong justification.

> ❝ Debugging is twice as hard as writing the code in the first place. Therefore, if you write the code as cleverly as possible, you are, by definition, not smart enough to debug it. ❞ -- Brian W. Kernighan

> ❝ Programs must be written for people to read, and only incidentally for machines to execute. ❞ -- Abelson and Sussman

Avoid premature optimizations

Optimizing code prematurely has several drawbacks:

- **You may not know which parts are the real performance bottlenecks**. This is especially the case when the code undergoes transformations (e.g., compiling, minifying, transpiling, etc.) before it becomes an executable. Ideally, you should use a profiler tool to identify the actual bottlenecks of the code first, and optimize only those parts.
- **Optimizing can complicate the code**, affecting correctness and readability.
- **Hand-optimized code can be harder for the compiler to optimize** (the simpler the code, the easier it is for the compiler to optimize). In many cases, a compiler can do a better job of optimizing the runtime code if you don't get in the way by trying to hand-optimize the source code.

Make it work, make it right, make it fast is popular saying in the industry, which means in most cases, getting the code to perform correctly should take priority over optimizing it. If the code doesn't work correctly, it has no value no matter how fast/efficient it is.

> ❝ Premature optimization is the root of all evil in programming. ❞ -- Donald Knuth

Of course, there are cases in which optimizing takes priority over other things e.g., when writing code for resource-constrained environments. This guideline is simply a caution that you should optimize only when it is really needed.

SLAP hard

Avoid having multiple levels of abstraction within a code fragment. Note: The book *The Productive Programmer* (by Neal Ford) calls this the *Single Level of Abstraction Principle* (SLAP) while the book *Clean Code* (by Robert C. Martin) calls this *One Level of Abstraction per Function*.

Bad (`readData();` and `salary = basic * rise + 1000;` are at different levels of abstraction)

```
readData();
salary = basic * rise + 1000;
tax = (taxable ? salary * 0.07 : 0);
displayResult();
```

Good (all statements are at the same level of abstraction)

```
readData();
processData();
displayResult();
```

Also ensure that the code is written at the highest level of abstraction possible.

Bad (all statements are at a low levels of abstraction)

```
low-level statement A1
low-level statement A2
low-level statement A3
low-level statement B1
low-level statement B2
if condition X :
    low-level statement C1
    low-level statement C2
```

Good (all statements are at the same high level of abstraction)

```
high-level step A
high-level step B
if condition X:
  high-level step C
```

That said, it is sometimes possible to pack *two* levels of abstraction into the code without affecting readability that much, provided each step in the higher-level logic is clearly marked using comments and separated (e.g., using a blank line) from adjacent steps.

Example: The following pseudocode has two levels of abstraction.

```
//high-level step A
low-level statement A1
low-level statement A2
low-level statement A3

//high-level step B
low-level statement B1
low-level statement B2

if condition X :
    //high-level step C
    low-level statement C1
    low-level statement C2
```

Advanced

Make the happy path prominent

The *happy path* should be clear and prominent in your code. Restructure the code to make the happy path (i.e., the execution path taken when everything goes well) less-nested as much as possible. It is the 'unusual' cases that should be nested. Someone reading the code should not get distracted by alternative paths taken when error conditions happen. One technique that could help in this regard is the use of guard clauses.

The following example shows how guard clauses can be used to reduce the nesting of the happy path.

Bad

```
if (!isUnusualCase) {  //detecting an unusual condition
    if (!isErrorCase) {
        start();    //main path
        process();
        cleanup();
        exit();
    } else {
        handleError();
    }
} else {
    handleUnusualCase(); //handling that unusual condition
}
```

In the code above,

- unusual condition detections are separated from their handling.
- the main path is nested deeply.

Good

```
if (isUnusualCase) { //Guard Clause
    handleUnusualCase();
    return;
}

if (isErrorCase) { //Guard Clause
    handleError();
    return;
}

start();
process();
cleanup();
exit();
```

In contrast, the above code

- deals with unusual conditions as soon as they are detected so that the reader doesn't have to remember them for long.
- keeps the main path un-indented.

The following pseudocode example shows how to reduce the nesting of the happy path inside a loop using a `continue` statement:

Bad

```
for (condition1)
    if (condition2)
        statement A
        statement B
        statement C
        statement D
statement E
```

Good

```
for (condition1)
    if (not condition2)
        continue
    statement A
    statement B
    statement C
    statement D
statement E
```

Guideline: Follow a standard

Introduction

One essential way to improve code quality is to follow a consistent style. That is why software engineers usually follow a strict coding standard (aka *style guide*).

The aim of a coding standard is to make the entire codebase look like it was written by one person. A coding standard is usually specific to a programming language and specifies guidelines such as the locations of opening and closing braces, indentation styles and naming styles (e.g., whether to use *Hungarian style*, *Pascal casing*, *Camel casing*, etc.). It is important that the whole team/company uses the same coding standard and that the standard is generally not inconsistent with typical industry practices. If a company's coding standard is very different from what is typically used in the industry, new recruits will take longer to get used to the company's coding style.

IDEs can help to enforce some parts of a coding standard e.g., indentation rules.

Exercises:

What is the recommended approach regarding coding standards?

What

Go through the Java coding standard at @SE-EDU and learn the *basic* style rules.

Exercises:

Find coding standard violations

Intermediate

Go through the Java coding standard at @SE-EDU and learn the *intermediate* style rules.

Exercises:

Unsuitable variable name

Find intermediate coding standard violations

Guideline: Name well

Introduction

Proper naming improves the readability of code. It also reduces bugs caused by ambiguities regarding the intent of a variable or a method.

> There are only two hard things in Computer Science: cache invalidation and naming things. -- Phil Karlton

Basic

Use nouns for things and verbs for actions

> Every system is built from a domain-specific language designed by the programmers to describe that system. Functions are the verbs of that language, and classes are the nouns.
>
> -- Robert C. Martin, *Clean Code: A Handbook of Agile Software Craftsmanship*

Use nouns for classes/variables and verbs for methods/functions.

Name for a	Bad	Good
Class	`CheckLimit`	`LimitChecker`
Method	`result()`	`calculate()`

Distinguish clearly between single-valued and multi-valued variables.

Good

```
Person student;
ArrayList<Person> students;
```

Use standard words

Use correct spelling in names. Avoid 'texting-style' spelling. **Avoid foreign language words, slang, and names that are only meaningful within specific contexts/times** e.g., terms from private jokes, a TV show currently popular in your country.

Intermediate

Use name to explain

A name is not just for differentiation; it should explain the named entity to the reader accurately and at a sufficient level of detail.

Bad	Good
`processInput()` (what 'process'?)	`removeWhiteSpaceFromInput()`
`flag`	`isValidInput`

Bad	Good
`temp`	

If a name has multiple words, they should be in a sensible order.

Bad	Good
`bySizeOrder()`	`orderBySize()`

Imagine going to the doctor's and saying "My eye1 is swollen"! Don't use numbers or case to distinguish names.

Bad	Bad	Good
`value1` , `value2`	`value` , `Value`	`originalValue` , `finalValue`

Not too long, not too short

While it is preferable not to have lengthy names, names that are 'too short' are even worse. If you must abbreviate or use acronyms, do it consistently. Explain their full meaning at an obvious location.

Avoid misleading names

Related things should be named similarly, while unrelated things should NOT.

Example: Consider these variables

- `colorBlack` : hex value for color black
- `colorWhite` : hex value for color white
- `colorBlue` : number of times blue is used
- `hexForRed` : hex value for color red

This is misleading because `colorBlue` is named similar to `colorWhite` and `colorBlack` but has a different purpose while `hexForRed` is named differently but has a very similar purpose to the first two variables. The following is better:

- `hexForBlack` `hexForWhite` `hexForRed`
- `blueColorCount`

Avoid misleading or ambiguous names (e.g., those with multiple meanings), similar sounding names, hard-to-pronounce ones (e.g., avoid ambiguities like "is that a lowercase L, capital I or number 1?", or "is that number 0 or letter O?"), almost similar names.

Bad	Good	Reason
`phase0`	`phaseZero`	Is that zero or letter O?
`rwrLgtDirn`	`rowerLegitDirection`	Hard to pronounce
`right` `left` `wrong`	`rightDirection` `leftDirection` `wrongResponse`	`right` is for 'correct' or 'opposite of 'left'?
`redBooks` `readBooks`	`redColorBooks` `booksRead`	`red` and `read` (past tense) sounds the same
`FiletMignon`	`egg`	If the requirement is just a name of a food, `egg` is a much easier to type/say choice than `FiletMignon`

Guideline: Avoid unsafe shortcuts

Introduction

It is safer to use language constructs in the way they are meant to be used, even if the language allows shortcuts. Such coding practices are common sources of bugs. Know them and avoid them.

Basic

Use the default branch

Always include a default branch in `case` statements. This ensures that all possible outcomes have been considered at the branching point.

Furthermore, use the `default` branch for the intended default action and not just to execute the last option. If there is no default action, you can use the `default` branch to detect errors (i.e., if execution reached the `default` branch, raise a suitable error). This also applies to the final `else` of an `if-else` construct. That is, the final `else` should mean 'everything else', and not the final option. Do not use `else` when an `if` condition can be explicitly specified, unless there is absolutely no other possibility.

Bad

```
if (red) print "red";
else print "blue";
```

Good

```
if (red) print "red";
else if (blue) print "blue";
else error("incorrect input");
```

Don't recycle variables or parameters

- **Use one variable for one purpose.** Do not reuse a variable for a different purpose other than its intended one, just because the data type is the same.
- **Do not *reuse* formal parameters as local variables** inside the method.

Bad

```
double computeRectangleArea(double length, double width) {
    length = length * width;  // parameter reused as a variable
    return length;
}
```

Good

```
double computeRectangleArea(double length, double width) {
    double area;
    area = length * width;
    return area;
}
```

Avoid empty catch blocks

Avoid empty `catch` statements, as they are a way to ignore errors silently (which is not a good thing). In cases when it is unavoidable, at least give a comment to explain why the `catch` block is left empty.

Delete dead code

Get rid of unused code the moment it becomes redundant. You might feel reluctant to delete code you have painstakingly written, even if you have no use for that code anymore ("I spent a lot of time writing that code; what if I need it again?"). Consider all code as baggage you have to carry. If you need that code again, simply recover it from the revision control tool you are using. Deleting code you wrote previously is a sign that you are improving.

Intermediate

Minimize scope of variables

Minimize global variables. Global variables may be the most convenient way to pass information around, but they do create implicit links between code segments that use the global variable. Avoid them as much as possible.

Define variables in the least possible scope. For example, if the variable is used only within the `if` block of the conditional statement, it should be declared inside that `if` block.

> **The most powerful technique for minimizing the scope of a local variable is to declare it where it is first used.** -- *Effective Java*, by Joshua Bloch

Resources:

- Refactoring: Reduce Scope of Variable

Minimize code duplication

Code duplication, especially when you copy-paste-modify code, often indicates a poor quality implementation. While it may not be possible to have zero duplication, always think twice before duplicating code; most often there is a better alternative.

This guideline is closely related to the DRY Principle.

Guideline: Comment minimally, but sufficiently

Introduction

> Good code is its own best documentation. As you're about to add a comment, ask yourself, 'How can I improve the code so that this comment isn't needed?' Improve the code and then document it to make it even clearer. -- Steve McConnell, Author of *Clean Code*

Some think commenting heavily increases the 'code quality'. That is not so. Avoid writing comments to explain bad code. Improve the code to make it self-explanatory.

Basic

Do not repeat the obvious

Do not repeat in comments information that is already obvious from the code. If the code is self-explanatory, a comment may not be needed.

Bad

```
//increment x
x++;

//trim the input
trimInput();
```

Write to the reader

Write comments targeting other programmers reading the code. Do not write comments as if they are private notes to yourself. Instead, One type of comment that is almost always useful is the *header comment* that you write for a class or an operation to explain its purpose.

Bad Reason: this comment will only make sense to the person who wrote it

```
// a quick trim function used to fix bug I detected overnight
void trimInput() {
    ....
}
```

Good

```
/** Trims the input of leading and trailing spaces */
void trimInput() {
    ....
}
```

Intermediate

Explain WHAT and WHY, not HOW

Comments should explain the *WHAT* and *WHY* aspects of the code, rather than the *HOW* aspect.

✓ **WHAT: The specification of what the code is *supposed* to do.** The reader can compare such comments to the implementation to verify if the implementation is correct.

Example: This method is possibly buggy because the implementation does not seem to match the comment. In this case, the comment could help the reader to detect the bug.

```
/** Removes all spaces from the {@code input} */
void compact(String input) {
    input.trim();
}
```

✓ **WHY: The rationale for the current implementation.**

Example: Without this comment, the reader will not know the reason for calling this method.

```
// Remove spaces to comply with IE23.5 formatting rules
compact(input);
```

✗ **HOW: The explanation for how the code works.** This should already be apparent from the code, if the code is self-explanatory. Adding comments to explain the same thing is redundant.

Example:

Bad Reason: Comment explains how the code works.

```
// return true if both left end and right end are correct
//     or the size has not incremented
return (left && right) || (input.size() == size);
```

Good Reason: The code is now self-explanatory -- the comment is no longer needed.

```
boolean isSameSize = (input.size() == size);
return (isLeftEndCorrect && isRightEndCorrect) || isSameSize;
```